

# Technical Data Sheet

Silicone Rubber Sheeting General Purpose Solid



## Material

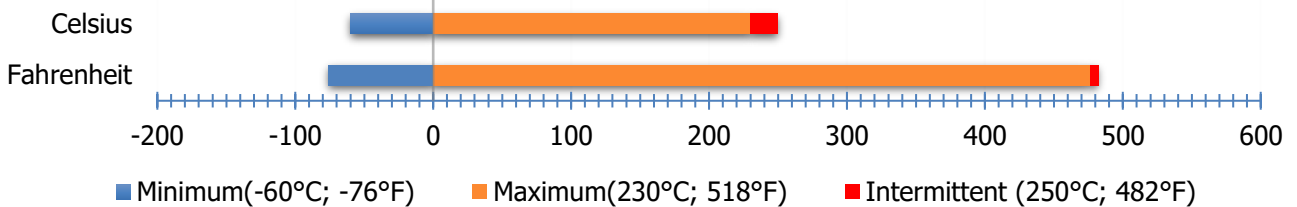
General Purpose Silicone Rubber



## Available Grades

GP 30, GP 40, GP 50, GP 60, GP 70, GP 80

## Temperature Range



## General Information

Our colour range has been tested to and is in accordance with the American Food and Drugs Administration (FDA) 21 CFR 177-2600, BFR XV Empfehlung and colour bleed test according to BFR Empfehlung part B 11 & Directive EC 1935/2004.

Translucent, Blue, White, Red and Black have been approved by the WRAS (Water Regulations Advisory Service) for use in contact with potable water at temperatures up to 85°C (185°F).

These products meet the flammability requirements of FAR 25/JAR 25/CS 25 Appendix F, Part 1, (a)(1)(iv) and (a)(1)(v) horizontal flammability test and Automotive Standard PART 571MVSS302.

## Environmental Resistance

Silicone rubber products have an excellent resistance to:

- Ozone
- Oxidation
- Ultraviolet light
- Corona discharge
- Cosmic radiation
- Ionising radiation
- Weathering in general

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**SILEX LTD**

Silex Ltd, Unit 10, Oakhanger Farm Business Park, Oakhanger, Bordon, Hampshire, GU35 9JA. Tel 00 44 1420 470360, Fax 00 44 1420 472802  
E-Mail: [sales@silex.co.uk](mailto:sales@silex.co.uk), Website: [www.silex.co.uk](http://www.silex.co.uk), Registered company number: 1951973. Place of registration: England & Wales.

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## Availability Format

- Supplied in continuous roll lengths
- Widths of 1000mm and 1200mm (standard), 1000mm x 1000mm for 15mm+
- 1500mm and 1800mm on request
- Pressure sensitive adhesive backing
- Punched and water jet gaskets
- Slit strips
- Full range of standard colours
- Capability to colour match

## Typical Applications

- Automotive
- Domestic & Commercial
- Catering
- Electronics
- Construction
- Energy
- Food & Beverage
- Heating and Ventilation (HVAC)
- Industrial
- Insulation
- Lighting
- Marine

## Mechanical Properties

|                                |                   | GP20          | GP 30         | GP 40         | GP 50         | GP 60          | GP 70          | GP 80         |                                                          |
|--------------------------------|-------------------|---------------|---------------|---------------|---------------|----------------|----------------|---------------|----------------------------------------------------------|
| Property                       | Units             | Typical Value | Typical Value | Typical Value | Typical Value | Typical Value  | Typical Value  | Typical Value | Test Method                                              |
| Hardness                       | Shore A<br>±5     | 20            | 30            | 40            | 50            | 60             | 70             | 80            | ASTM D2240<br>DIN ISO 7619-1                             |
| Tensile Strength               | Mpa<br>(PSI)      | 10,1          | 8.0<br>(1150) | 9.0<br>(1300) | 9.0<br>(1300) | 10.0<br>(1450) | 10.0<br>(1450) | 7.0<br>(1015) | ISO 37 type 1<br>ASTM D412 die C<br>DIN 53504<br>type S1 |
| Elongation to Failure          | %                 | 979           | 500           | 525           | 450           | 400            | 320            | 215           | ISO 37 type 1<br>ASTM D412 die C<br>DIN 53504<br>type S1 |
| Tear Strength                  | N/mm<br>(lb./in.) | 25,0          | 16.0<br>(91)  | 23<br>(130)   | 18<br>(100)   | 22<br>(125)    | 20<br>(115)    | 17<br>(88.5)  | ASTM D624 die B                                          |
| Compression Set 24 Hrs @ 150°C | %                 | 23            | 23            | 25            | 15            | 13             | 9              | 18            | BS 903 pt A6<br>type B<br>DIN/ISO 815 type B             |
| Compression Set 22 Hrs @ 300°F | %                 | -             | 20            | 22            | 16            | 10             | 10             | 16            | ASTM D395 method B type 2                                |

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## General Characteristics

| Test                  | Result                                                | Standard       |
|-----------------------|-------------------------------------------------------|----------------|
| Brittle Point         | -80°C (-112 °F)                                       | ASTM D746      |
| Limiting Oxygen Index | 24.0 %                                                | BS 2782 Part 1 |
| Thermal Conductivity  | 0.24 W.m <sup>-1</sup> .K <sup>-1</sup>               | VDE 0304       |
| Radiation Resistance  | >10 <sup>5</sup> Grays (10 <sup>7</sup> Rads) typical |                |
| Dielectric Strength   | 23 kV.mm <sup>-1</sup>                                | VDE 0303       |
| Dielectric Constant   | 2.9                                                   | VDE 0303       |
| Dissipation Factor    | 3x10 <sup>-4</sup>                                    | VDE 0303       |
| Volume Resistivity    | 3x10 <sup>15</sup> Ω.cm                               | VDE 0303       |

## Accreditations

- FDA 21 CFR 177-2600 compliant
- BfR compliant
- EU 1935/2004
- WRAS compliant
- FAR 25/JAR 25/CS 25 Appendix F, Part 1, (a)(1)(iv) (a)(1)(v) horizontal flammability test
- Automotive Standard PART 571MVSS302
- REACH compliant and ROHS compliant

## Additional Information

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